

denai — Detailed AI Phase Matrix

Exact AI Responsibilities, Model Usage, Current Status & Execution Roles

This document defines the exact operational role of every AI system used in the denai roadmap. For every phase, the document specifies: • current completion status • exact Claude model usage • whether Sonnet uses Thinking mode or standard execution • exact Gemini Pro responsibilities • optional DeepSeek responsibilities • escalation conditions for Opus • execution boundaries and constraints

PHASE 0 — Original Platform Foundation

STATUS: COMPLETED

Purpose: Build the original deterministic clinical platform. Completed Systems: • calcAI() • calcAIMulti() • ClinicalEngine • patient management • report generation • What-If simulator • PWA support • accessibility systems Claude Usage: • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - clinical workflow reasoning - architecture direction - deterministic scoring reasoning - treatment logic planning Gemini Pro Usage: Responsibilities: - roadmap organization - architecture summaries - document formatting - long-context consolidation DeepSeek: Not required in this phase. Opus: Not required.

PHASE 1 — UX Hardening & Stability Preparation

STATUS: COMPLETED

Purpose: Stabilize frontend behavior and remove UX inconsistencies. Completed Work: • rendering stabilization • accessibility refinement • edge-case handling • deterministic UI behavior Claude Usage: • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - rendering consistency reasoning - UX stability planning - accessibility analysis - edge-case architecture reasoning Gemini Pro Usage: Responsibilities: - UX audit summarization - issue categorization - roadmap organization DeepSeek Usage (Optional): Responsibilities: - brainstorming hidden edge cases - identifying strange UI failure scenarios Opus: Not required.

PHASE 2 — Regression Infrastructure

STATUS: COMPLETED

Purpose: Build deterministic regression and rollback infrastructure. Completed Work: • DandyTests stabilization • golden snapshots • regression harness • rollback philosophy • parity validation system Claude Usage: • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - regression strategy - parity methodology - rollback sequencing - deterministic validation logic Gemini Pro Usage: Responsibilities: - regression documentation synthesis - test categorization - audit summaries DeepSeek Usage (Optional): Responsibilities: - regression blind-spot criticism - hidden regression brainstorming Opus: Not required.

PHASE 3 — Architectural Intelligence

STATUS: MOSTLY COMPLETED

Purpose: Generate full extraction intelligence before modularization. Completed: • Extraction Matrix • Dependency Intelligence • Shared State Registry • Orchestration Mapping • Hidden Behavioral Contracts • Deterministic Extraction Constraints Remaining: • dom_lifecycle_audit.md • synchronization_analysis.md • clinical_engine_extraction.md Claude Usage: • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - dependency reasoning - synchronization analysis - orchestration mapping - extraction safety reasoning - render-order analysis - state coupling analysis Gemini Pro Usage: Responsibilities: - cross-artifact consistency checking - identifying roadmap inconsistencies - large-context synthesis - architecture document consolidation DeepSeek Usage (Optional): Responsibilities: - adversarial architecture criticism - blind-spot detection - extraction-risk criticism Opus: Allowed ONLY if: - synchronization paradox appears - orchestration deadlock appears - architecture conflict cannot be resolved

PHASE 4 — Wave Execution Playbooks

STATUS: PENDING

Purpose: Create exact extraction sequencing for all modularization waves. Planned Work: • rollback gates • validation checkpoints • extraction ordering • forbidden extraction combinations • synchronization-safe sequencing Claude Usage: • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - wave sequencing - extraction strategy - rollback planning - deterministic execution planning - risk ordering Gemini Pro Usage: Responsibilities: - playbook organization - consistency review - execution matrix formatting - cross-wave dependency summaries DeepSeek Usage (Optional): Responsibilities: - criticize extraction order - identify unsafe sequencing Opus: Only if severe orchestration conflict appears.

PHASE 5 — Claude Code Environment Preparation

STATUS: PENDING

Purpose: Prepare safe execution environment for modularization. Planned Work: • Git workflow • diff review workflow • sandbox extraction practice • rollback checkpoints • Claude Code onboarding Claude Usage: • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - teaching workflow philosophy - explaining safe modularization process - review-before-apply methodology • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - execution practice - sandbox implementation - filesystem operations - small extraction exercises Gemini Pro Usage: Responsibilities: - learning summaries - workflow organization - command reference structuring DeepSeek: Not required. Opus: Not required.

PHASE 6 — Wave 1 Modularization

STATUS: NOT STARTED

Purpose: Extract LOW-RISK pure utilities. Targets: • constants • validators • geometry helpers • escapeHtml() • CleanupRegistry • DOMPurify wrappers Claude Usage: • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - extraction execution - imports/exports - file creation - utility extraction - regression repair • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - extraction oversight - diff review guidance - architectural validation Gemini Pro Usage: Responsibilities: - extraction tracking - documentation synthesis - wave progress summaries DeepSeek Usage (Optional): Responsibilities: - identify hidden utility coupling Opus: Not required.

PHASE 7 — Wave 2 Clinical Engine Extraction

STATUS: NOT STARTED

Purpose: Extract deterministic clinical AI engine. Targets: • calcAI() • calcAIMulti() • ClinicalEngine • scoring logic • recommendation systems Claude Usage: • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - extraction execution - parity fixes - module implementation - imports/exports • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - recommendation parity reasoning - scoring validation - extraction oversight - architecture continuity Gemini Pro Usage: Responsibilities: - parity documentation - cross-test synthesis - architecture summaries DeepSeek Usage (Optional): Responsibilities: - edge-case criticism - recommendation drift criticism Opus: Allowed ONLY if: - recommendation determinism breaks - engine orchestration paradox appears

PHASE 8 — Wave 3 State & Reports Extraction

STATUS: NOT STARTED

Purpose: Extract persistence and state systems safely. Targets: • persistence • patient management • reports • tempState • history systems Claude Usage: • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - state extraction - persistence migration - report extraction - regression fixes • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - shared-state reasoning - mutation-order analysis - persistence safety planning Gemini Pro Usage: Responsibilities: - state dependency summaries - persistence documentation - report structure synthesis DeepSeek Usage (Optional): Responsibilities: - identify hidden state coupling Opus: Allowed ONLY if: - severe state synchronization conflict appears

PHASE 9 — Wave 4 Rendering Systems

STATUS: NOT STARTED

Purpose: Extract high-risk rendering systems. Targets: • AI card rendering • comparison rendering • SVG rendering • renderComparisonTable() • renderRisk() Claude Usage: • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - rendering extraction - DOM-safe refactors - render module implementation - regression fixes • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - DOM lifecycle reasoning - render-order analysis - orchestration oversight - synchronization-safe extraction reasoning Gemini Pro Usage: Responsibilities: - render dependency synthesis - lifecycle documentation - extraction summaries DeepSeek Usage (Optional): Responsibilities: - stale DOM criticism - render-risk auditing Opus: Allowed ONLY if: - orchestration deadlock appears - severe render synchronization paradox appears

PHASE 10 — Wave 5 Orchestrator & Synchronization

STATUS: NOT STARTED

Purpose: Extract highest-risk orchestration systems. Targets: • render orchestration • renderMainPanels() • debounce chains • event systems • synchronization logic Claude Usage: • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - orchestrator extraction - synchronization-safe implementation - event migration - regression repair • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - synchronization reasoning - orchestration continuity - render coordination analysis - timing analysis Gemini Pro Usage: Responsibilities: - orchestration documentation - synchronization summaries - cross-wave continuity tracking DeepSeek Usage (Optional): Responsibilities: - synchronization criticism - orchestration blind-spot detection Opus: Allowed ONLY for: - severe synchronization paradoxes - orchestration collapse risks

PHASE 11 — Post-Modular Stabilization

STATUS: NOT STARTED

Purpose: Stabilize architecture after modularization. Planned Work: • regression sweeps • render timing validation • synchronization validation • import stabilization Claude Usage: • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - stabilization fixes - cleanup - dependency repair - regression debugging • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - architecture review - stabilization reasoning - cleanup prioritization Gemini Pro Usage: Responsibilities: - stabilization reporting - cleanup summaries - architecture consistency synthesis DeepSeek: Optional regression criticism. Opus: Not normally required.

PHASE 12 — Developer Infrastructure

STATUS: NOT STARTED

Purpose: Modernize engineering infrastructure. Planned Work: • Vite • Vitest • Playwright • GitHub Actions • CI/CD Claude Usage: • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - infrastructure implementation - build tooling - test setup - CI implementation • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - infrastructure planning - migration sequencing - tooling architecture Gemini Pro Usage: Responsibilities: - infrastructure documentation - CI summaries - deployment organization DeepSeek: Optional deployment-risk criticism. Opus: Only for major infrastructure redesign dilemmas.

PHASE 13 — SaaS Operationalization

STATUS: NOT STARTED

Purpose: Transform platform into SaaS system. Planned Work: • backend APIs • PostgreSQL • authentication • clinic architecture • cloud reports Claude Usage: • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - backend implementation - database integration - auth implementation - API development • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - SaaS architecture reasoning - multi-user workflow planning - permissions architecture - backend sequencing Gemini Pro Usage: Responsibilities: - system documentation - architecture summaries - API documentation synthesis DeepSeek Usage (Optional): Responsibilities: - security criticism - multi-tenant blind-spot review Opus: Allowed ONLY for: - complex multi-tenant architecture dilemmas

PHASE 14 — Clinical Validation & Scaling

STATUS: NOT STARTED

Purpose: Validate platform clinically and scale commercially. Planned Work: • real dentist testing • UX refinement • telemetry • analytics • monetization systems Claude Usage: • Claude Code — Sonnet 4.6 STANDARD Responsibilities: - telemetry implementation - analytics instrumentation - dashboard implementation • Claude Chat — Sonnet 4.6 WITH THINKING Responsibilities: - workflow analysis - UX reasoning - scaling strategy - monetization planning Gemini Pro Usage: Responsibilities: - analytics synthesis - roadmap reporting - scaling summaries DeepSeek: Optional adoption-risk criticism. Opus: Allowed ONLY for: - large-scale platform pivots

FINAL AI RESPONSIBILITY MATRIX

AI System	Primary Usage
Claude Chat — Sonnet 4.6 WITH THINKING	architecture, orchestration, planning, reasoning
Claude Code — Sonnet 4.6 STANDARD	execution, implementation, modularization
Gemini Pro	synthesis, documentation, consistency review
DeepSeek	skeptical auditing & blind-spot criticism
Claude Opus	emergency escalation only

The denai roadmap is intentionally designed around deterministic preservation, incremental extraction, orchestration safety, regression discipline, and controlled AI role separation.